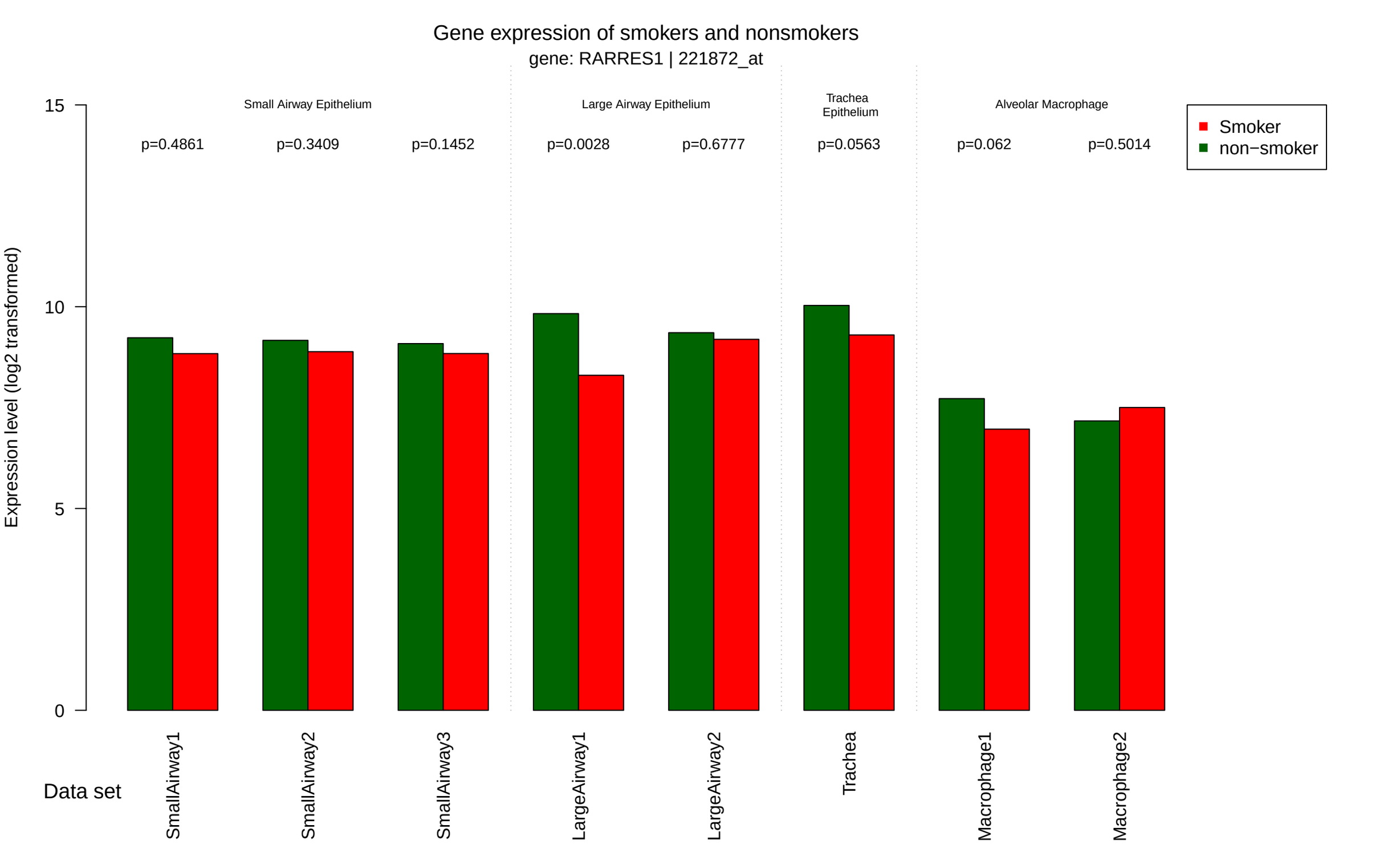
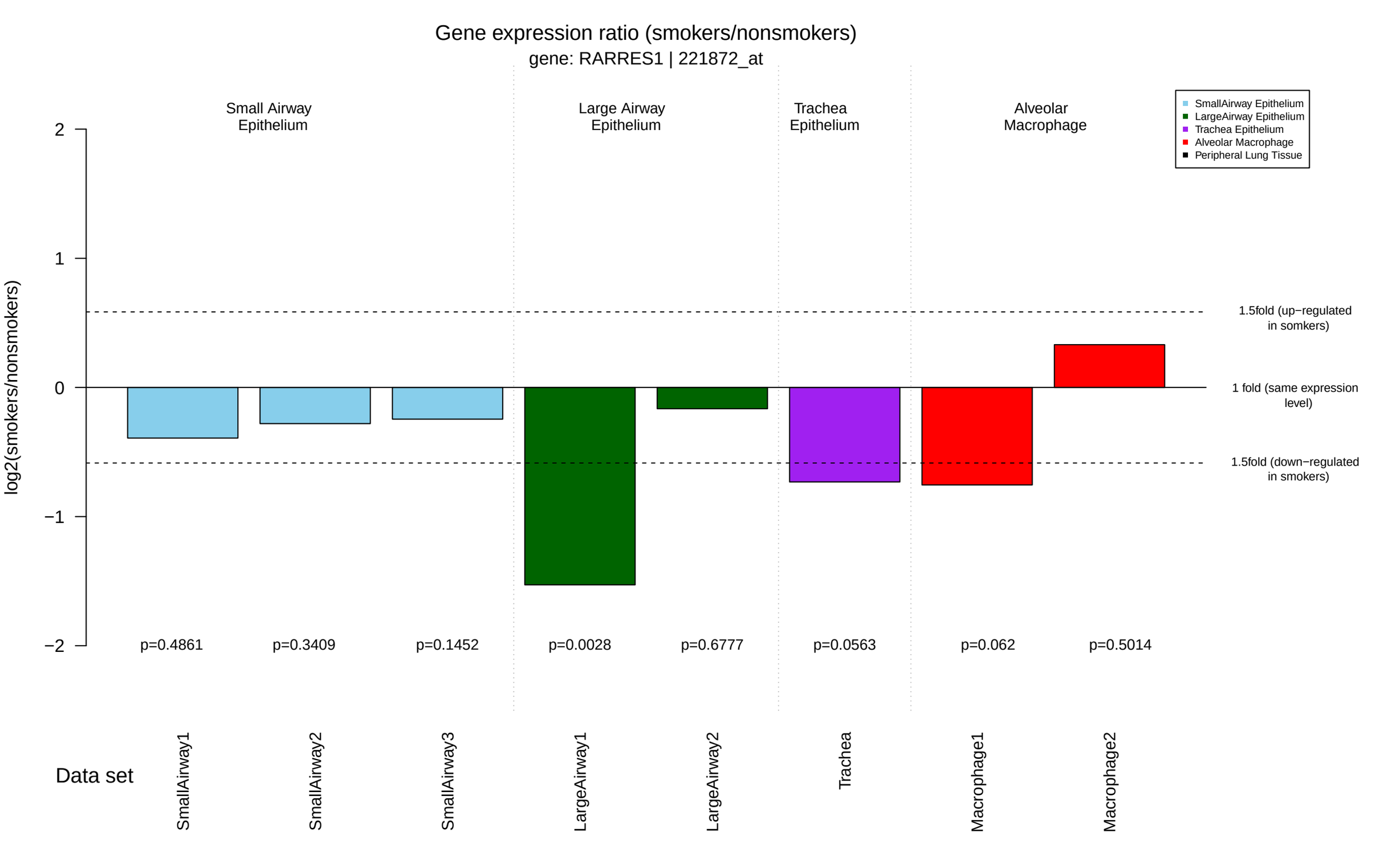
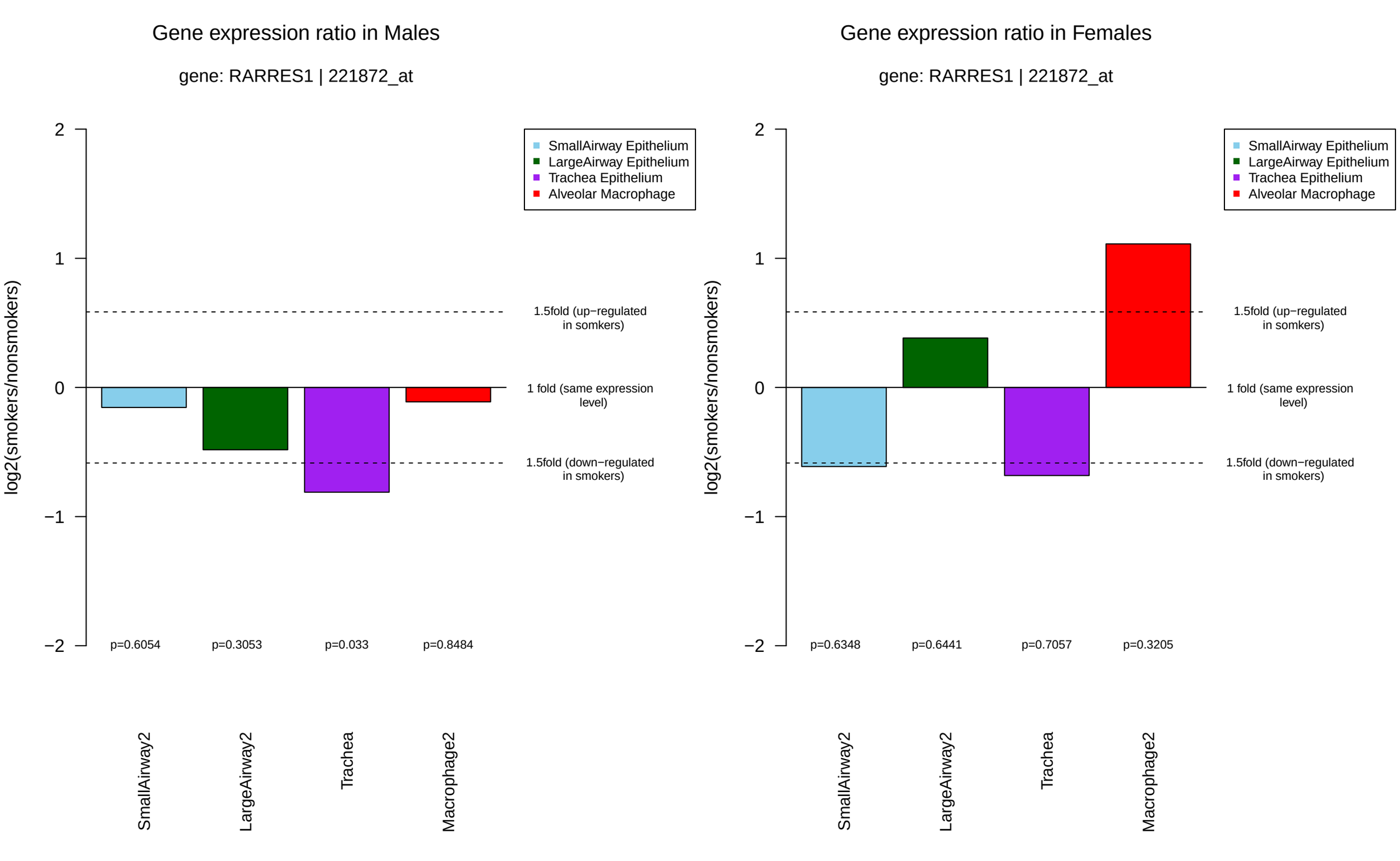
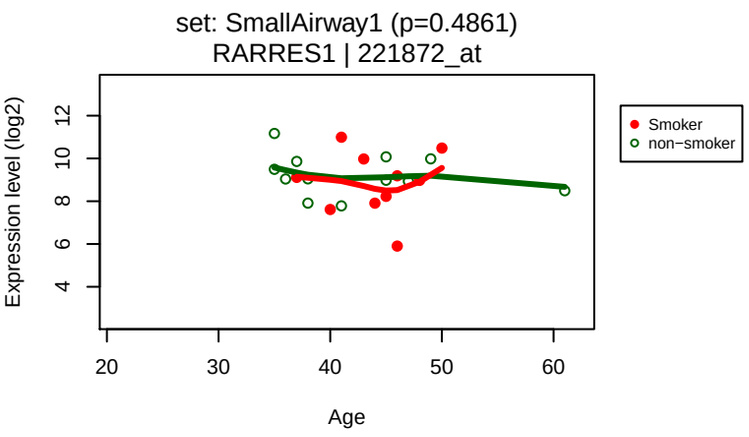
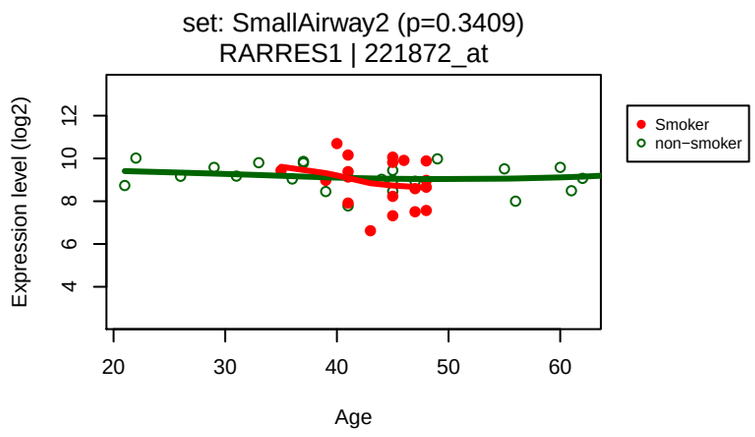
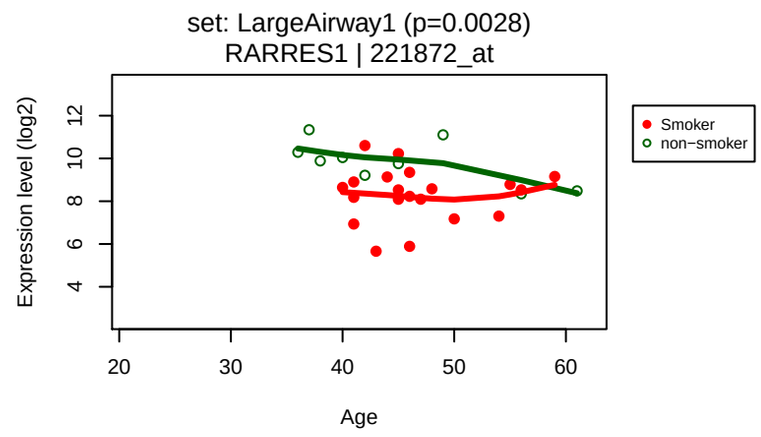

Gene expression of smokers and nonsmokers
gene: RARRES1 | 221872_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4861
p=0.3409
p=0.1452
p=0.0028
p=0.6777
p=0.0563
p=0.062
p=0.5014
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

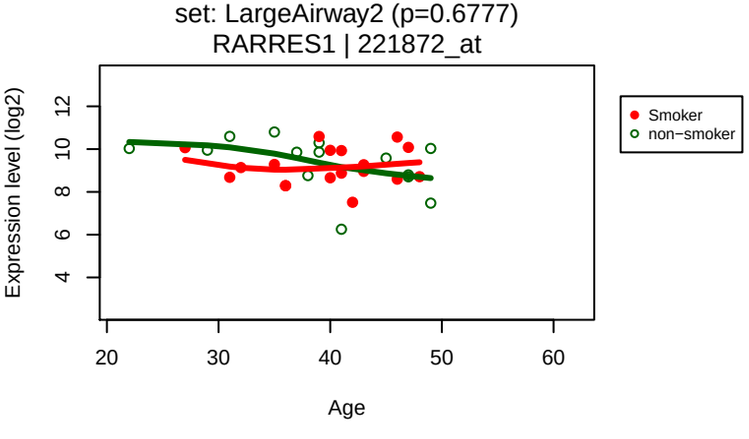

Gene expression ratio (smokers/nonsmokers)
gene: RARRES1 | 221872_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4861
p=0.3409
p=0.1452
p=0.0028
p=0.6777
p=0.0563
p=0.062
p=0.5014
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

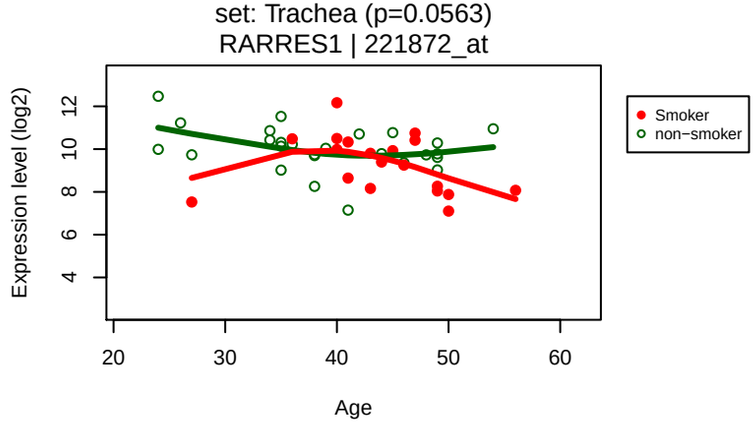

Gene expression ratio in Males
gene: RARRES1 | 221872_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6054
p=0.3053
p=0.033
p=0.8484
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RARRES1 | 221872_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6348
p=0.6441
p=0.7057
p=0.3205
SmallAirway2
LargeAirway2
Trachea
Macrophage2

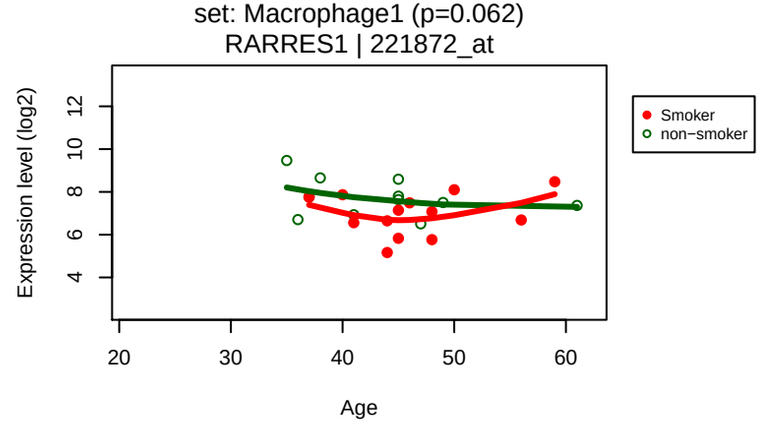

set: SmallAirway1 (p=0.4861)
RARRES1 | 221872_at
Expression level (log2)
Age
Smoker
non−smoker

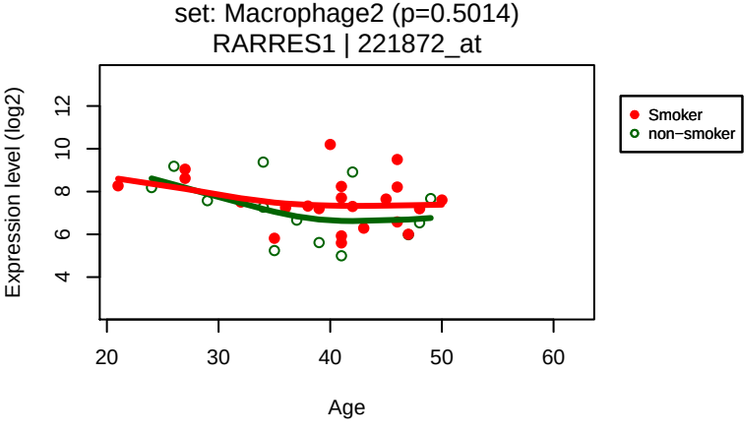

set: SmallAirway2 (p=0.3409)
RARRES1 | 221872_at
Expression level (log2)
Age
Smoker
non−smoker

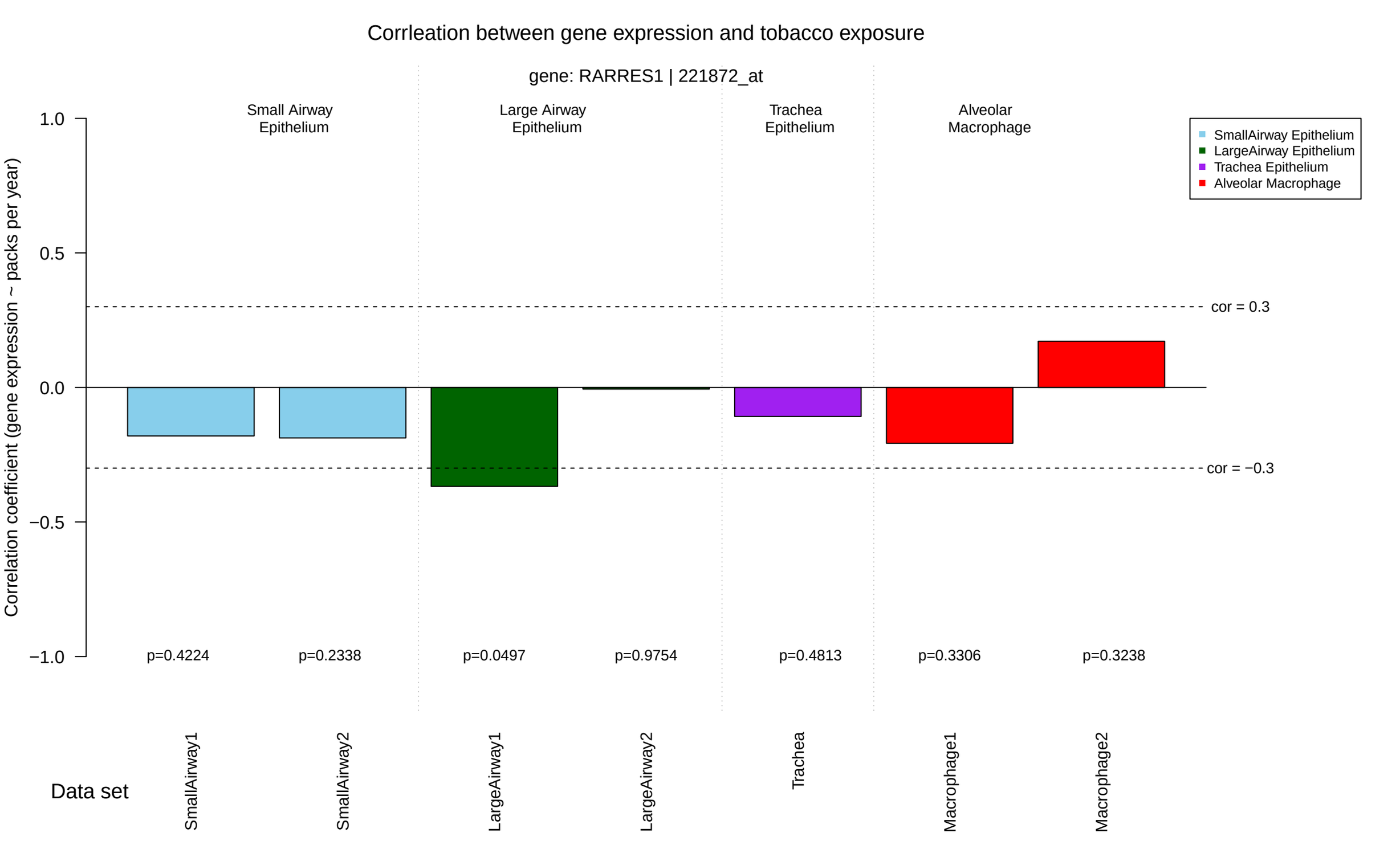

set: LargeAirway1 (p=0.0028)
RARRES1 | 221872_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6777)
RARRES1 | 221872_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0563)
RARRES1 | 221872_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.062)
RARRES1 | 221872_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5014)
RARRES1 | 221872_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RARRES1 | 221872_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4224
p=0.2338
p=0.0497
p=0.9754
p=0.4813
p=0.3306
p=0.3238
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2